

DIGITAL DATA, ANALOG SIGNALS

The most familiar use of this transformation is for transmitting digital data through the public telephone network. The telephone network was designed to receive, switch, and transmit analog signals in the voice-frequency range of about 300 to 3400 Hz. It is not at present suitable for handling digital signals from the subscriber locations. Thus digital devices are attached to the network via a modem (modulator-demodulator), which converts digital data to analog signals, and vice versa.

For the telephone network, modems are used that produce signals in the voice-frequency range. The same basic techniques are used for modems that produce signals at higher frequencies (e.g., microwave).

Modulation involves operation on one or more of the three characteristics of a carrier signal: amplitude, frequency, and phase. Accordingly, there are three basic encoding or modulation techniques for transforming digital data into analog signals, as illustrated in Figure 5.7: amplitude shift keying (ASK), frequency shift keying (FSK), and phase shift keying (PSK). In all these cases, the resulting signal occupies a bandwidth centered on the carrier frequency.

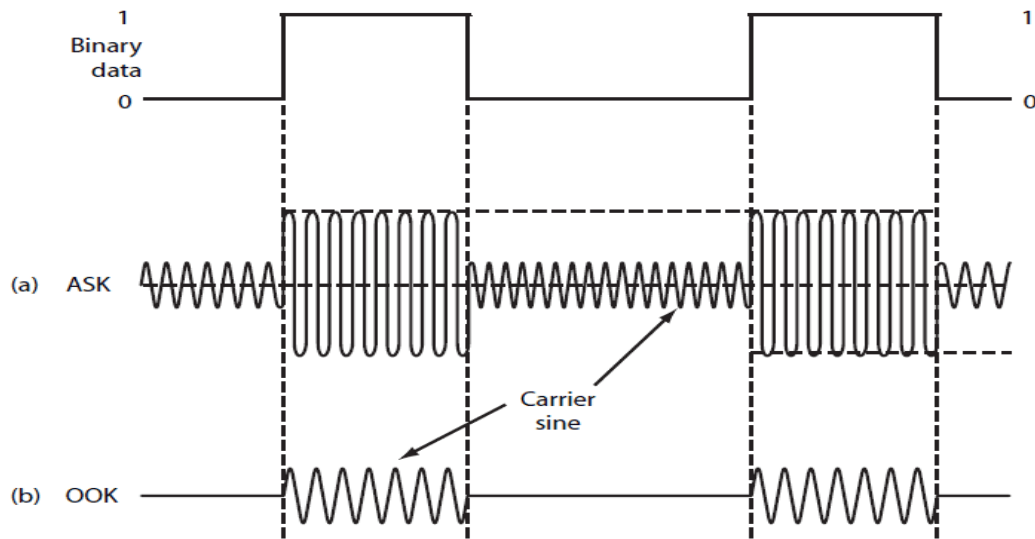
Amplitude Shift Keying

In ASK, the two binary values are represented by two different amplitudes of the carrier frequency. Commonly, one of the amplitudes is zero; that is, one binary digit is represented by the presence, at constant amplitude, of the carrier, the other by the absence of the carrier. The resulting transmitted signal for one-bit time is where the carrier signal is ASK is susceptible to sudden gain changes and is a rather inefficient modulation technique. On voice-grade lines, it is typically used only up to 1200 bps.

$$s(t) = \begin{cases} A \cos 2\pi f t & \text{Binary 1} \\ 0 & \text{Binary 0} \end{cases} \quad 1$$

The ASK technique is used to transmit digital data over optical fiber. For LED (light-emitting diode) transmitters, Equation (5.2) is valid. That is, one signal element is

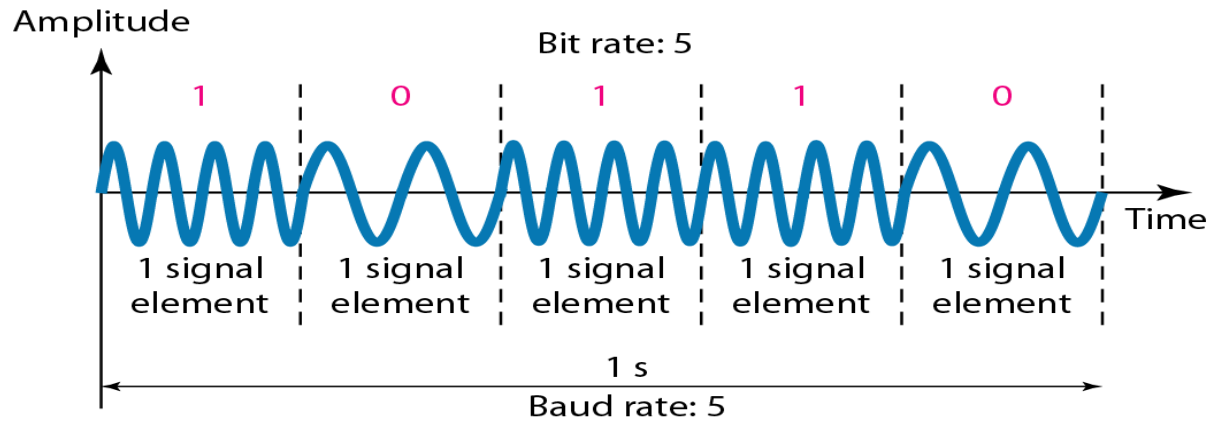
represented by a light pulse while the other signal element is represented by the absence of light. Laser transmitters normally have a fixed “bias” current that causes the device to emit a low light level. This low level represents one signal element, while a higher-amplitude lightwave represents another signal element.



Frequency Shift Keying (FSK)

The most common form of FSK is binary FSK (BFSK), in which the two binary values are represented by two different frequencies near the carrier frequency. The resulting transmitted signal for one-bit time is

$$s(t) = \begin{cases} A\cos(2\pi f_1 t) & \text{Binary 1} \\ A\cos(2\pi f_0 t) & \text{Binary 0} \end{cases}$$



BFSK is less susceptible to error than ASK. On voice-grade lines, it is typically used up to 1200 bps. It is also commonly used for high-frequency (3 to 30 MHz) radio transmission. It can also be used at even higher frequencies on local area networks that use coaxial cable.

Phase Shift Keying

In PSK, the phase of the carrier signal is shifted to represent data. Two-Level PSK The simplest scheme uses two phases to represent the two binary digits and is known as binary phase shift keying. The resulting transmitted signal for one bit time is

$$s(t) = \begin{cases} A \cos(2\pi f_c t) & \text{Binary 0} \\ A \cos(2\pi f_c t + \pi) = -A \cos(2\pi f_c t) & \text{Binary 1} \end{cases}$$

INFORMATION CAPACITY, BITS, BIT RATE, BAUD, AND M-ARY ENCODING

$$C \propto B \times t$$

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From Equation 1, it can be seen that information capacity is a linear function of bandwidth and transmission time and is directly proportional to both.

2-2-2 M-ary Encoding

M-ary is a term derived from the word *binary*. *M* simply represents a digit that corresponds to the number of conditions, levels, or combinations possible for a given number of binary variables.

For example, a digital signal with four possible conditions (voltage levels, frequencies, phases, and so on) is an *M*-ary system where $M = 4$. If there are eight possible conditions, $M = 8$ and so forth. The number of bits necessary to produce a given number of conditions is expressed mathematically as

$$N = \log_2 M \quad 2$$

where N = number of bits necessary

M = number of conditions, levels, or combinations possible with N bits

Equation 3 can be simplified and rearranged to express the number of conditions possible with N bits as

$$2^N = M \quad 3$$

For example, with one bit, only $2^1 = 2$ conditions are possible. With two bits, $2^2 = 4$ conditions are possible, with three bits, $2^3 = 8$ conditions are possible, and so on.

2-2-3 Baud and Minimum Bandwidth

Baud refers to the rate of change of a signal on the transmission medium after encoding and modulation have occurred. Hence, baud is a unit of transmission rate, modulation rate, or symbol rate and, therefore, the terms symbols per second and baud are often used interchangeably.

Mathematically, baud is the reciprocal of the time of one output signaling element, and a signaling element may represent several information bits. Baud is expressed as

$$\text{Baud} = \frac{1}{t_s} \quad 4$$

Baud = symbol rate (baud per second) t_s = time of one signaling element (seconds) . The minimum theoretical bandwidth

The minimum theoretical bandwidth necessary to propagate a signal is called the minimum Nyquist bandwidth or sometimes the minimum Nyquist frequency.

Thus, $C = 2B$, where C is the bit rate in bps and B is the ideal Nyquist bandwidth.

The relationship between bandwidth and bit rate also applies to the opposite situation. For a given bandwidth (B), the highest theoretical bit rate is $2B$.

For example, a standard telephone circuit has a bandwidth of approximately 2700 Hz, which has the capacity to propagate 5400 bps through it. However, if more than two levels are used for signaling (higher-than-binary encoding), more than one bit may be transmitted at a time, and it is possible to propagate a bit rate that exceeds 2B.

Using multilevel signaling, the Nyquist formulation for channel capacity is

$$C = 2B \log_2 M \quad 5$$

C is the capacity in bits per second, B is the frequency bandwidth (Minimum Nyquist bandwidth) in Hertz, and M is the number of levels a single symbol can take on. This "idealized" capacity equation shows us that data rate is proportional to twice the bandwidth and logarithmically proportional to M. It is considered "idealized" since environmental influences, particularly noise, is not considered.

The maximum data rate capacity of a digital communications system is a factor of the noise environment, frequency bandwidth, and modulation scheme. Nyquist, Shannon, and Hartley who were pioneers in the science of information technology, discovered equations that took these factors into consideration. Today, their work helps us to better understand how these factors influence one another. The Nyquist formula above provided a relationship between capacity and bandwidth under idealized conditions where noise is not considered.

Shannon-Hartley developed a similar equation for capacity; however they included the Signal-to-Noise ratio (SNR) which provides a slightly more realistic answer. Shannon-Hartley's equation provides the "theoretical" maximum capacity for a signal given its frequency bandwidth and SNR. Typically, digital communication systems do not reach this theoretical capacity due to many factors such as modulation scheme and the overall noise environment. However, it remains the goal of many

communications engineers to get as close to the theoretical maximum capacity as possible in designing their systems.

Mathematically stated, *the Shannon limit_for information capacity is*

$$C = B \log_2 \left(1 + \frac{S}{N} \right) = 3.32 \log_{10} \left(1 + \frac{S}{N} \right) \quad 6$$

where C = information capacity (bps) B = bandwidth (hertz) $\frac{S}{N}$ = signal-to-noise power ratio (unitless)

For a standard telephone circuit with a signal-to-noise power ratio of 1000 (30 dB) and a bandwidth of 2.7 kHz, the Shannon limit for information capacity is $I = (3.32)(2700) \log_{10} (1 + 1000) = 26.9$ kbps

Shannon's formula is often misunderstood. The results of the preceding example indicate that 26.9 kbps can be propagated through a 2.7-kHz communications channel. This may be true, but it cannot be done with a binary system. To achieve an information transmission rate of 26.9 kbps through a 2.7-kHz channel, each symbol transmitted must contain more than one bit.

The maximum rate at which data can be transmitted over a given communication path, or channel, under given conditions, is referred to as the **channel capacity**.

There are four concepts here that we are trying to relate to one another.

- **Data rate**: The rate, in bits per second (bps), at which data can be communicated
- **Bandwidth**: The bandwidth of the transmitted signal as constrained by the transmitter and the nature of the transmission medium, expressed in cycles per second, or Hertz
- **Noise**: The average level of noise over the communications path
- **Error rate**: The rate at which errors occur, where an error is the reception of a 1 when a 0 was transmitted or the reception of a 0 when a 1 was transmitted

Performance of the digital modulation schemes

In looking at the performance of various digital-to-analog modulation schemes, the first parameter of interest is the bandwidth of the modulated signal. This depends on a variety of factors, including the definition of bandwidth used and the filtering technique used to create the bandpass signal. The transmission bandwidth for ASK is of the form of equation 7

$$ASK \quad B = (1 + r)R$$

where R is the bit rate and r is related to the technique by which the signal is filtered to establish a bandwidth for transmission; typically $0 < r < 1$. Thus the bandwidth is directly related to the bit rate. The preceding formula is also valid for PSK and, under certain assumptions, FSK. With multilevel PSK (MPSK) significant improvements in bandwidth can be achieved. In general,

$$MPSK \quad B_T = \left(\frac{1 + r}{L} \right) R = \left(\frac{1 + r}{\log_2 M} \right) R$$

where L is the number of bits encoded per signal element and M is the number of different signal elements. For multilevel FSK (MFSK), we have

$$MFSK \quad B_T = \left(\frac{(1 + r)M}{L} \right) R = \left(\frac{(1 + r)M}{\log_2 M} \right) R$$

Bandwidth efficiency as the name suggests, is a parameter that measures the efficiency with which bandwidth can be used to transmit data. It is the ratio of the data rate to the transmission bandwidth.

TRANSMISSION LINE IMPAIRMENTS

With any communications system, the signal that is received may differ from the signal that is transmitted due to various transmission impairments. For analog signals, these impairments can degrade the signal quality. For digital signals, bit errors may be introduced, such that a binary 1 is transformed into a binary 0 or vice versa.

The most significant impairments are

- Attenuation and attenuation distortion
- Delay distortion
- Noise

Attenuation

The strength of a signal falls off with distance over any transmission medium. For guided media, this reduction in strength, or attenuation, is generally exponential and thus is typically expressed as a constant number of decibels per unit distance. For unguided media, attenuation is a more complex function of distance and the makeup of the atmosphere. Attenuation introduces three considerations for the transmission engineer.

- First, a received signal must have sufficient strength so that the electronic circuitry in the receiver can detect the signal.
- Second, the signal must maintain a level sufficiently higher than noise to be received without error.
- Third, attenuation varies with frequency.

Delay Distortion

Delay distortion occurs because the velocity of propagation of a signal through a guided medium varies with frequency. For a bandlimited signal, the velocity tends to be highest near the center frequency and fall off toward the two edges of the band.

Thus various frequency components of a signal will arrive at the receiver at different times, resulting in phase shifts between the different frequencies. This effect is referred to as delay distortion because the received signal is distorted due to varying delays experienced at its constituent frequencies. Delay distortion is particularly critical for digital data. Consider that a sequence of bits is being transmitted, using either analog or digital signals. Because of delay distortion, some of the signal components of one-bit position will spill over into other bit positions, causing intersymbol interference, which is a major limitation to maximum bit rate over a transmission channel. Equalizing techniques can also be used for delay distortion.

Noise

For any data transmission event, the received signal will consist of the transmitted signal, modified by the various distortions imposed by the transmission system, plus additional unwanted signals that are inserted somewhere between transmission and reception. The latter, undesired signals are referred to as noise. Noise is the major limiting factor in communications system performance. Noise may be divided into four categories

- Thermal noise
- Intermodulation noise
- Crosstalk
- Impulse noise